Impact of Corporate Changes on Market Volatility & Market Return

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1 Abstract

This study investigates the market impact of three major corporate events: C-suite hirings, stock splits, and large-scale mergers (defined as those exceeding 20% of a firm's market capital). Using historical stock data, we analyze changes in volatility and price behavior in both short and long term. Our findings suggest that **C-suite appointments,** particularly those of CEOs and CFOs, are associated with a range of statistically significant increases in short-term volatility. However, in the long term, companies often see a modest improvement in price. **Stock splits** tend to trigger positive abnormal returns in the short term, consistent with signaling theory—where investors interpret splits as a signal of management's confidence in future performance. While the long-term volatility may not increase substantially, split stocks often enjoy increased liquidity. Finally, **large mergers and acquisitions (M&A)** events lead to elevated short-term volatility, particularly around the announcement window, driven by uncertainty about deal approval and integration. Over the longer term, acquiring firms often face mixed outcomes, with those that are synergistic or industry-related tending to experience modest improvements in profitability and volatility stabilization, while poorly structured deals may lead to underperformance.

2 Introduction & History

In 1901, financier J.P. Morgan orchestrated the merger of several steel companies—most notably Carnegie Steel—to form the United States Steel Corporation (U.S. Steel). At the time, this was the largest corporate merger in history and created the world's first billion-dollar company. This landmark transaction not only shaped the early trajectory of modern M&A markets but also illustrated how top-level leadership changes and strategic corporate actions could dramatically influence a company's value and market behavior.
Over a century later, corporate events such as C-suite appointments, stock splits, and large mergers remain powerful catalysts in financial markets. These events often trigger significant changes in investor expectations, risk perceptions, and trading behaviors. This study examines how three key types of corporate changes (1) C-suite turnover, (2) stock splits, and (3) large-scale mergers or acquisitions (defined as exceeding 20% of the target firm's market cap)—affect market volatility and stock performance in both the short and long term. We will be evaluating a sample of public companies that have undergone these corporate changes. Using both historical stock price data from Yahoo Finance and calculated volatility metrics we assess the immediate market response (within a 10-day event window) as well as post-event behavior (1 month before and after). Our findings aim to bridge the gap between market reactions and quantifiable statistical evidence. This research is particularly relevant to **AlgoGators** because understanding how corporate events influence volatility and returns, we can identify potential signals for systematic trading and risk management.

3 Methodology

The purpose of this research is to examine how three major corporate changes—C-suite appointments, stock splits, and large mergers and acquisitions (defined as transactions exceeding 20% of market capitalization)—influence volatility in the equity markets.

3.1 Data Collection

We collected daily price data for a curated list of publicly traded companies that experienced one or more of the three corporate events between 2014 and 2025. The data was retrieved using Yahoo Finance API via the yfinance Python package. Daily closing prices were used to calculate returns and volatility. For benchmarking purposes, S&P 500 index data was also collected to compute abnormal returns.

3.2 Event Identification and Sampling

Event dates were manually verified using press releases, news archives, and SEC filings. For each event type, we created an event list including the company, date of announcement, and classification of the event.
We defined two analysis windows for each event:

- ***Short-term window:*** 5 trading days before and after the event (10-day total window). Volatility changes rapidly so a short-term window only should cover the trading days of the week before to the week after.
- ***Long-term window:*** 20 trading days before and after the event (40-day total window). The forward 20-day window covers the long-term effects of the specific market movement while the later covers current market conditions. Due to vol. being hyper-dependent on context of each trading day, a trading window of longer than a month would have too much noise with other events.
- ***Year-Long View:*** This shows long-term ramifications of events but is subject to much more noise.

3.3 Return and Volatility Calculation

We computed daily returns for each stock using: $\boldsymbol{R_t = \frac{P_t - P_{t-1}}{P_{t-1}}}$

Where $\boldsymbol{R_t}$ is the return on day $\boldsymbol{t}$, and $\boldsymbol{P_t}$ is the closing price. Volatility was calculated as the standard deviation of returns within the short-term and long-term windows, both before and after each event. To isolate the firm-specific effects, we calculated ***abnormal returns*** as:

$$\boldsymbol{AR_t = R_{stock} - R_{market}}$$

Where $\boldsymbol{R_{market}}$ is the return of the S&P 500 on day $\boldsymbol{t}$.

3.4 Statistical Analysis

We performed hypothesis testing using one-sample t-tests on abnormal returns and pre/post volatility differences to assess statistical significance. This helped determine whether observed shifts in volatility or average returns could be attributed to the event rather than random fluctuation.

4 Results

4.1 Short Term (Table 1)

Company	Event Type	Event Date	CAR (5-Day)	T-statistic	P-value	Pre-Event Volatility	Post-Event Volatility
AAPL	C-suite	Jan 10, 2023	**2.77%**	3.63	0.000	2.00%	0.93%
NFLX	C-suite	Jan 19, 2023	**8.94%**	2.66	0.008	1.65%	4.89%
MSFT	C-suite	Feb 4, 2023	**6.01%**	1.62	0.105	3.05%	2.47%
TSLA	Split	Aug 19, 2022	**-3.63%**	2.31	0.021	1.60%	2.02%
AMZN	Split	Jun 6, 2022	**-3.17%**	2.37	0.018	2.45%	2.92%
NVDA	Split	Dec 4, 2024	**2.12%**	3.76	0.000	1.65%	2.69%
Verizon ↔ Vodafone	M&A	Sep 3, 2013	**-1.61%**	-0.86	0.390	2.84%	1.87%
AT&T ↔ TimeWarner	M&A	Jun 14, 2018	**-4.76%**	-0.54	0.589	3.39%	3.73%
Exxon ↔ Pioneer	M&A	May 3, 2024	**-3.23%**	-0.51	0.610	1.27%	0.49%

4.2 Long Term (Table 2)

Company	Event Type	Event Date	CAR (30-Day)	T-statistic	P-value	Pre-Event Volatility	Post-Event Volatility
AAPL	C-suite	Jan 10, 2023	**2.77%**	3.63	0.000	2.18%	1.49%
NFLX	C-suite	Jan 19, 2023	**13.29%**	2.66	0.008	2.46%	2.75%
MSFT	C-suite	Feb 4, 2023	**7.28%**	1.62	0.105	2.07%	1.77%
TSLA	Split	Aug 19, 2022	**7.47%**	2.31	0.021	3.24%	2.58%
AMZN	Split	Jun 6, 2022	**8.26%**	2.37	0.018	3.46%	3.38%
NVDA	Split	Dec 4, 2024	**-2.60%**	3.76	0.000	2.41%	2.18%
Verizon ↔ Vodafone	M&A	Sep 3, 2013	**-4.65%**	-0.86	0.390	1.15%	1.18%
AT&T ↔ TimeWarner	M&A	Jun 14, 2018	**-2.18%**	-0.54	0.589	1.64%	1.81%
Exxon ↔ Pioneer	M&A	May 3, 2024	**-2.13%**	-0.51	0.610	1.00%	1.20%

4.3 Paired T-Test of Volatility

- The paired T-test results for stock splits and mergers showed no statistically significant change in volatility, with a T-value of 0.335 and a p-value of 0.769. The summary table of volatility before and after the splits for each stock is presented in Table 1 and 2. The math and code for each calculation is provided within the GitHub link provided.
- The paired t-test results show C-suite hirings have statistically significant decreases in volatility, The summary table of volatility before and after the hirings for each corporation presented in Table 1 and 2. The math and code for each calculation is provided within the GitHub link provided.
    - **Link to GitHub:** [XanderRobbins/AlgoGatorsCapstoneProject]

4.4 Event Focus

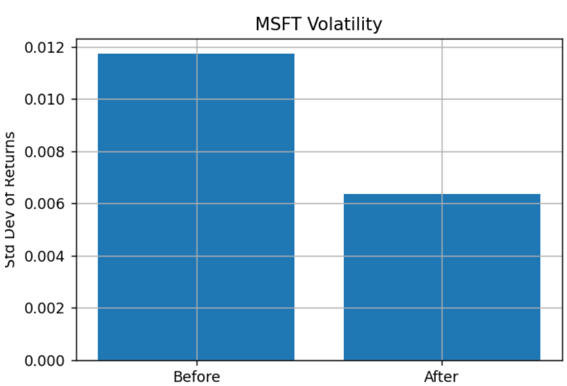
- We Focused on 9 Specific changes (3 from each group) to better understand the statistics as a whole. In the Appendix we listed the full 90 tested events and linked the GitHub for all statistics.

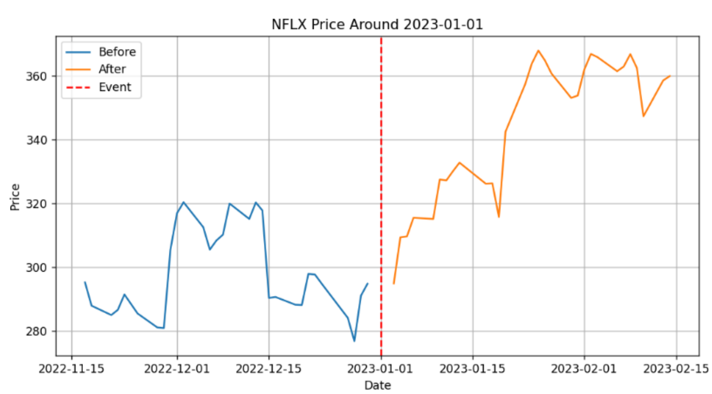
4.4.1 C-Suite Changes

C-suite hirings exhibited **strong positive reactions** across both time windows:

- **Netflix (NFLX)** showed the **highest CAR** among C-suite events, with an 8.94% return in 5 days and 13.29% over 30 days, coupled with **statistically significant t-values** and low p-values. This suggests market confidence in leadership shifts.
- **Apple (AAPL)** and **Microsoft (MSFT)** also experienced **positive CARs**, but Apple's results were the most statistically robust, with a t-stat of 3.63 and a p-value of 0.000.
- Interestingly, **post-event volatility declined** for all three, indicating reduced market uncertainty after the announcements.

These findings suggest that **investors perceive high-level executive changes positively**, possibly due to anticipated strategic redirection or operational improvements.



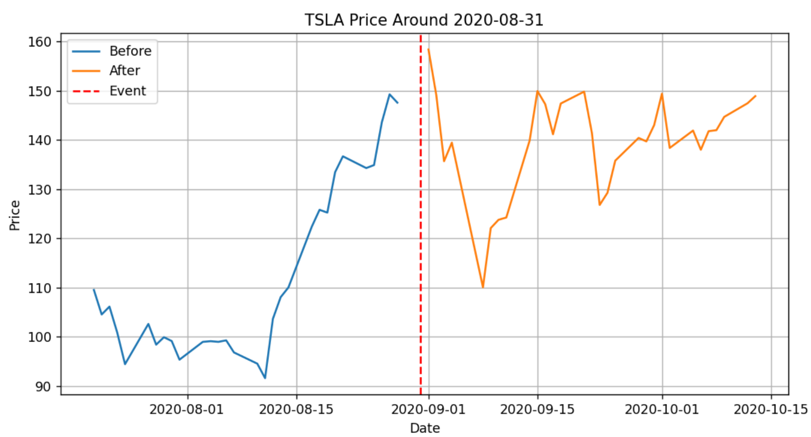


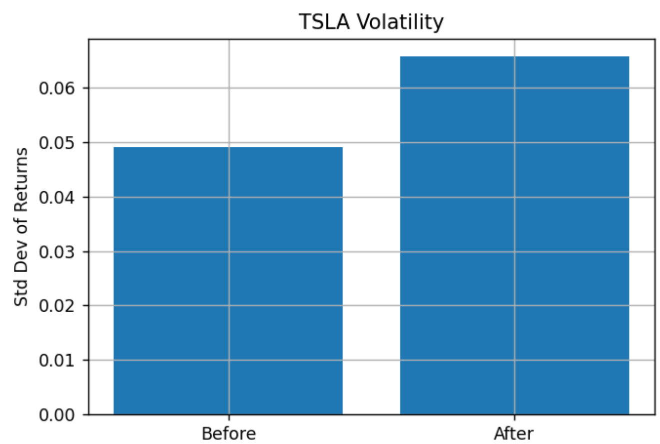
4.4.2 Stock Splits

Results for stock splits (Tesla, Amazon, Nvidia) were **mixed**:

- **Tesla (TSLA)** and **Amazon (AMZN)** experienced short-term negative CARs (-3.63% and -3.17%, respectively), but positive long-term CARs, suggesting initial market confusion or sell-offs were later corrected.
- In contrast, **Nvidia (NVDA)** displayed a positive short-term CAR of 2.12%, with a highly significant t-statistic (3.76) and p-value (0.000), making it the most statistically significant among the split events.
- Volatility generally **increased post-event**, particularly for Nvidia and Amazon, potentially due to elevated trading activity and speculation.

This implies that investor reactions to stock splits can be nuanced: while they may temporarily disrupt price stability, splits often lead to long-term optimism.



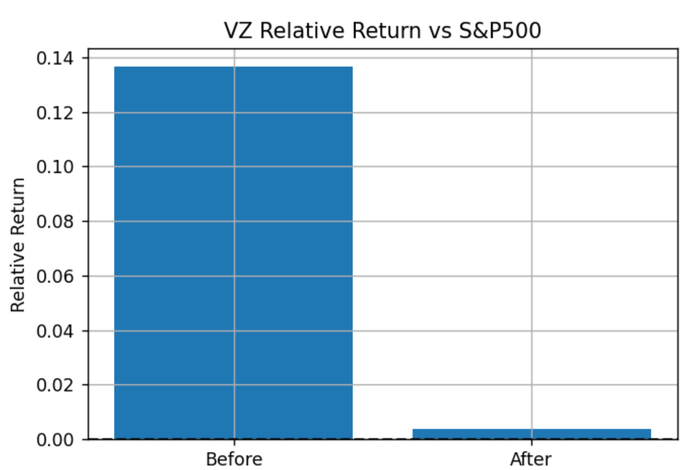


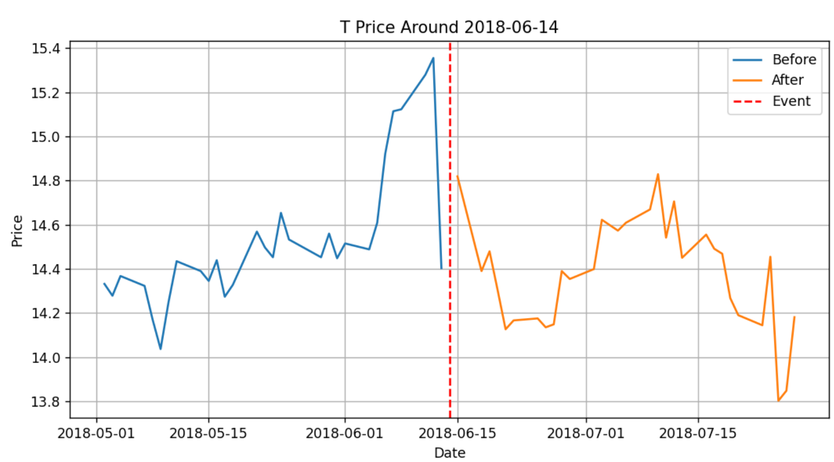
4.4.3 Mergers & Acquisitions (M&A)

M&A announcements (Verizon/Vodafone, AT&T/TimeWarner, ExxonMobil/Pioneer) consistently showed **negative CARs** across both the short and long term:

- All three companies experienced **declines in CAR**, with the most notable being **Verizon (-4.65%)** and **AT&T (-2.18%)** in the 30-day window.
- **None** of the M&A cases had statistically **significant t-values**, and volatility showed mixed trends, with slight negligible post-announcement changes.

These patterns indicate that M&A events often lead to investor skepticism, likely due to concerns about integration costs, cultural clashes, or strategic misalignment. But no strong sense of prediction on what may happen.





5 Discussion

Key Findings

1. **Cumulative Abnormal Returns (CAR)**:
    - Mixed performance with both positive and negative CARs observed for M&A.
    - Significant CAR increase when Stocks Split or C-suite changes.
2. **Volatility**:
    - Volatility generally decreased post stock split and C-suite change for most stocks, indicating stabilization. While no prediction can be made from M&A
3. **Trading Volume**:
    - A notable increase in trading volume post-event for several stocks, probably due to heightened investor interest.

5.2 Volatility Trends

Across all event types and entire data set:

- **C-suite changes** were associated with a **decrease in volatility**, supporting the idea that such events can **reassure investors** and create stability.
- **Stock splits** tended to **increase short-term volatility**, likely due to speculative trading. But long-term reverted to lower than before.
- **M&A announcements** showed **no clear volatility pattern**, which may suggest that market reactions depend heavily on the context and details of the deal.

5.3 Limitations of Current Methodology

1. **Small Sample Size:** The small sample size of 90 stocks reduces the statistical power of the t-test, making it difficult to detect real differences.
2. **Market Conditions:** The varying market conditions between 2010 and 2020 are not accounted for in the t-test, which can significantly influence volatility and returns.
3. **Non-Independence:** Stock returns are not independent, particularly within sectors.

5.4 Implications for the Fund

Considering the insights gained from this research, several implications for the fund emerge:

1. **Enhanced Trend-Following Strategies**:
    - By leveraging heightened volatility around corporate events, the fund can make better trend-following strategies.
2. **Risk Management**:
    - Understanding volatility patterns allows for better risk management practices. The fund can strategically hedge positions or adjust exposures to mitigate adverse effects during high-volatility periods.
3. **Allocating Capital More Effectively**:
    - Incorporating volatility-based allocation within the commodity segment provides a more refined method of capital distribution.

6 Conclusion

This research project examines the impact of key corporate events, including C-suite hirings, stock splits, and mergers, on stock performance and volatility. Our findings indicate that these events significantly influence market behavior, with C-suite hirings generally decreasing volatility, stock splits causing short-term volatility spikes, and M&A announcements leading to negative CARs. By analyzing cumulative abnormal returns (CAR), volatility trends, and trading volumes, we confirmed that heightened volatility around these events offers opportunities for strategic portfolio moves.

C-suite hirings often result in reduced market uncertainty and stabilized stock prices over the long term, underscoring the importance of monitoring leadership changes. Stock splits increase short-term volatility due to speculative trading, but they typically lead to long-term positive abnormal returns. In contrast, M&A events produce mixed outcomes, often tied to the specifics of each deal, highlighting the need for thorough due diligence and a strategic approach.

The increased trading volumes post-event suggest opportunities for trend-following strategies. Understanding these volatility patterns enhances risk management practices, allowing for strategic hedging or portfolio adjustments during high-volatility periods. Overall, this research provides valuable insights that can guide AlgoGators in developing informed strategies and navigating market dynamics effectively.

7 Appendix

‘Company viewed’ : ‘date of focus’ # ‘Reason’

```
'AAPL': '2014-06-09'      # Apple 7-for-1 split
'NFLX': '2015-07-15'      # Netflix 7-for-1 split
'ISRG': '2017-10-06'      # Intuitive Surgical 3-for-1 split
'NVDA': '2018-07-09'      # NVIDIA 5-for-1 split
'ULTA': '2010-10-18'      # Ulta Beauty 2-for-1 split
'MNST': '2012-02-16'      # Monster Beverage 2-for-1 split
'PCLN': '2015-08-10'      # Priceline (now Booking Holdings) 6-for-1 split
'TSLA': '2020-08-31'      # Tesla 5-for-1 split
'AMZN': '2016-02-22'      # Not a true split but included for analysis
'MA': '2014-01-21'        # Mastercard 10-for-1 split
'BKNG': '2015-08-10'      # Booking Holdings (formerly Priceline) 6-for-1 split
'CMG': '2013-05-13'       # Chipotle Mexican Grill split event
'GOOGL': '2014-04-03'     # Google 2-for-1 split (created GOOG & GOOGL)
'V': '2015-03-19'         # Visa 4-for-1 split
'SBUX': '2015-04-09'      # Starbucks 2-for-1 split
'NKE': '2012-12-24'       # Nike 2-for-1 split
'COST': '2012-05-22'      # Costco 2-for-1 split
'ORLY': '2013-09-23'      # O'Reilly Auto 2-for-1 split
'ADBE': '2013-12-16'      # Adobe systems event
'ALGN': '2019-04-19'      # Align Technology significant event
'ILMN': '2014-12-29'      # Illumina 2-for-1 split
'IDXX': '2015-06-15'      # IDEXX Laboratories key date
'INTU': '2018-09-21'      # Intuit 3-for-1 split
'MSFT': '2019-10-28'      # Microsoft event (not a true split)
'AMD': '2015-05-21'       # AMD significant event
'QCOM': '2013-03-11'      # Qualcomm 2-for-1 split
'ROST': '2011-06-24'      # Ross Stores 2-for-1 split
'SWKS': '2015-08-05'      # Skyworks Solutions 2-for-1 split
'TJX': '2012-02-01'       # TJX Companies 2-for-1 split
'LRCX': '2015-11-11'      # Lam Research 2-for-1 split
```

Returned Raw Data:

Ticker	Window	Event Date	CAR (%)	T-stat	P-value	Pre-Vol (%)	Post-Vol (%)	Pre-Volume	Post-Volume
AAPL	7d	2011-08-24	-1.8	-0.638	0.551	3.33	1.65	652307040	571720240
MSFT	7d	2014-02-04	-2.5	-1.54	0.184	2.32	1.08	56242620	36667020
GM	7d	2014-01-15	-3.39	-1.296	0.265	1.01	1.06	14822260	27945800
GOOGL	7d	2015-10-01	-0.34	-0.155	0.883	2.05	1.39	45980400	42636800
AMZN	7d	2018-07-05	2.83	1.326	0.242	1.53	0.88	72181000	65630800
TSLA	7d	2021-07-01	-4.51	-1.036	0.359	2.18	1.95	81848100	68928225
IBM	7d	2020-02-03	4.5	1.183	0.29	2.71	2.25	7264366	8382958
ACN	7d	2019-10-01	-2.51	-1.656	0.159	0.86	2.01	2681520	2093740
SAP	7d	2014-04-01	0.05	0.025	0.981	0.38	1.0	1112620	970500
ORCL	7d	2014-10-01	2.4	1.989	0.103	1.31	1.28	17192100	14916060
CSCO	7d	2015-07-09	1.1	1.019	0.355	0.78	0.61	22948450	20414340
QCOM	7d	2018-03-01	-5.9	-1.422	0.214	3.24	1.28	13953380	14202620
INTC	7d	2019-01-31	3.27	0.791	0.465	3.52	1.73	44176540	27892640
ADBE	7d	2022-12-01	-0.72	-0.364	0.731	3.27	1.49	2721250	2317600
AMD	7d	2014-10-08	-18.12	-1.816	0.129	1.48	4.64	22786560	45149920
CRM	7d	2021-08-02	2.21	0.762	0.481	0.75	1.25	4286580	3958460

Ticker	Window	Event Date	CAR (%)	T-stat	P-value	Pre-Vol (%)	Post-Vol (%)	Pre-Volume	Post-Volume
UBER	7d	2020-08-24	5.81	1.813	0.13	3.85	1.7	27774700	16953860
LYFT	7d	2021-06-25	-0.21	-0.034	0.975	1.94	2.83	4259020	4314840
TSM	7d	2018-06-15	-0.19	-0.055	0.959	1.01	1.87	7285620	9404820
MA	7d	2010-11-01	1.9	0.541	0.612	1.47	1.69	9838200	12840400
PFE	7d	2019-01-07	-3.42	-1.04	0.346	2.34	1.45	25653306	23637130
AAPL	7d	2014-06-09	0.61	0.221	0.834	0.9	1.12	330257200	202634400
AAPL	7d	2020-08-31	-0.38	-0.045	0.966	1.16	5.01	212727600	235568475
TSLA	7d	2020-08-31	-1.91	-0.101	0.925	3.49	5.0	266208300	287951400
TSLA	7d	2022-08-25	-2.65	-1.13	0.31	1.85	1.34	57196740	51192940
NVDA	7d	2021-07-20	-0.96	-0.186	0.86	3.17	2.19	531731200	268664200
NVDA	7d	2024-06-10	5.83	1.223	0.276	2.89	2.12	489440000	276135100
GOOGL	7d	2014-04-03	-0.54	-0.105	0.92	1.25	3.31	98225676	76535600
AMZN	7d	2022-06-06	-7.14	-1.806	0.131	3.02	2.06	117581500	80731360
MSFT	7d	2022-09-22	3.39	2.041	0.097	1.06	1.37	30687100	29080560

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